

Two Spring Damper System

P.11 Robot Motion, Simulation, Optimization

Suraj Dayma March 22, 2026

1. Intro

System animation, fixed point analysis and clustering, and linear stability analysis for the **Two Spring and Two Damper System** with drag.

2. Problem and Dynamics:

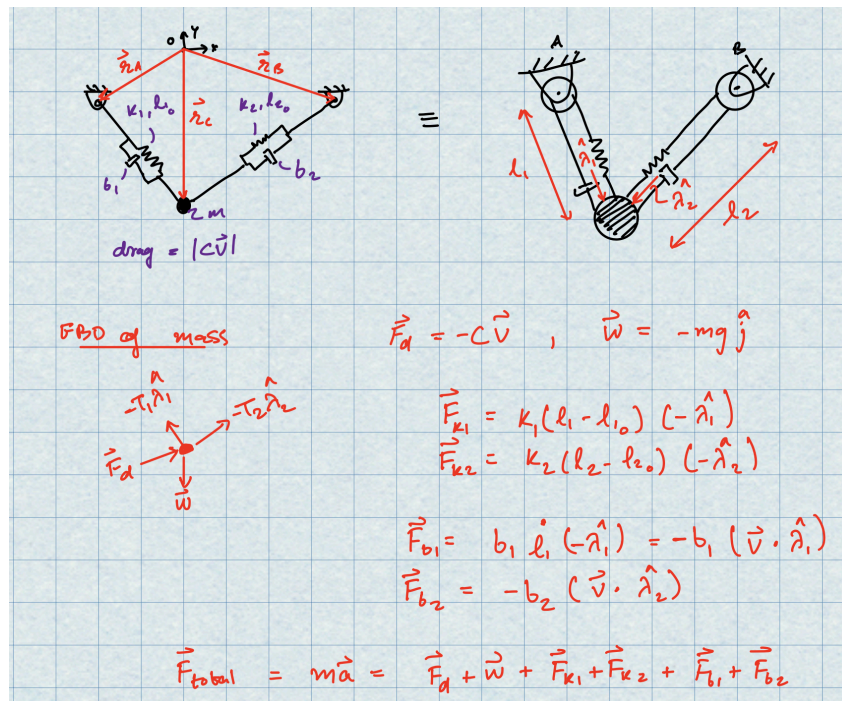


Figure 1: Force Balance

3. Fixed Points

The fixed points were found using `fsolve`, and clustering of fixed points was performed by using the following algorithms:

1. KMeans: within cluster sum of squares was found by varying K and the best K was determined by the elbow method.
2. DBScan: the distance parameter was kept very small (0.005) which converged correctly for every scenario I tried (except the infinite fixed point cases).

3.1. Fixed Points = 2

```
% System Constants
p.m = 1;
p.g = 10;

p.b1 = 1;
p.b2 = 1;

p.d = 0.6; % drag coef

p.k1 = 1;
p.k2 = 10;
p.l1_0 = 1;
p.l2_0 = 1;

p.r1 = [-1; 0];
p.r2 = [1; 0];
```

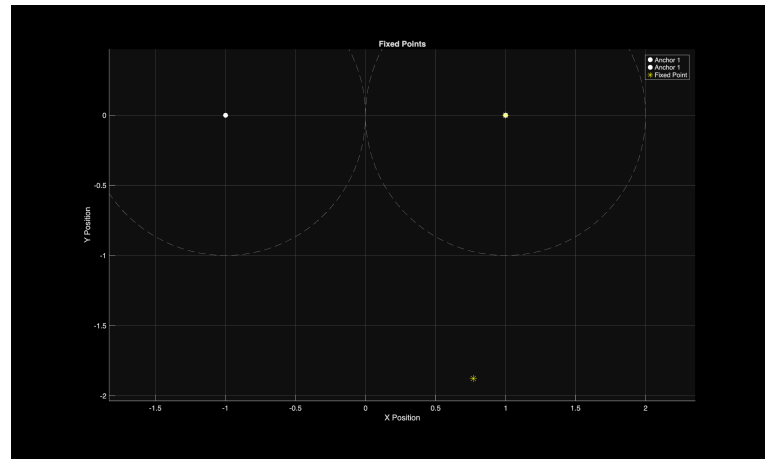


Figure 2: Fixed Points

Both KMeans and DBScan provided the right answer for this.

3.2. Fixed Points = 5

```
% System Constants
p.m = 1;
p.g = 0;

p.b1 = 1;
p.b2 = 1;

p.d = 1; % drag coef

p.k1 = 10;
p.k2 = 10;
p.l1_0 = 1.5;
p.l2_0 = 1.5;

p.r1 = [-1; 0];
p.r2 = [1; 0];
```

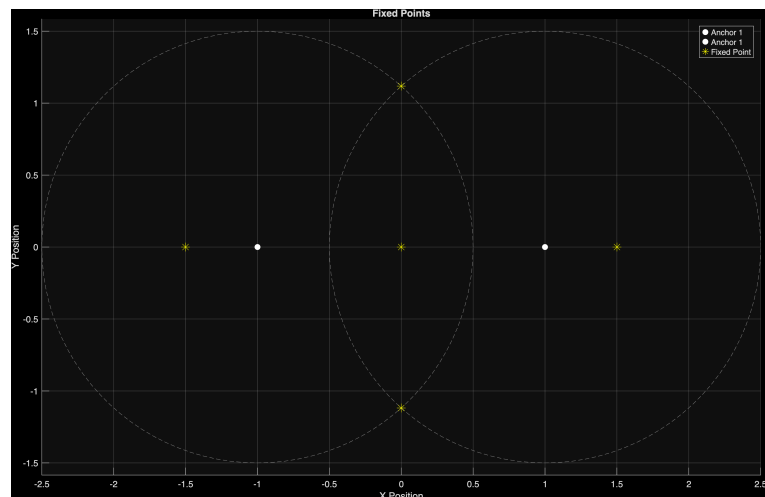


Figure 3: Fixed Points

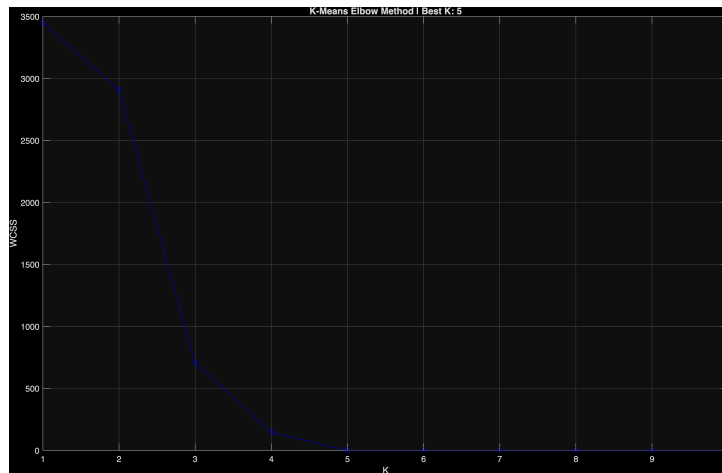


Figure 4: KMean - Best K = 3

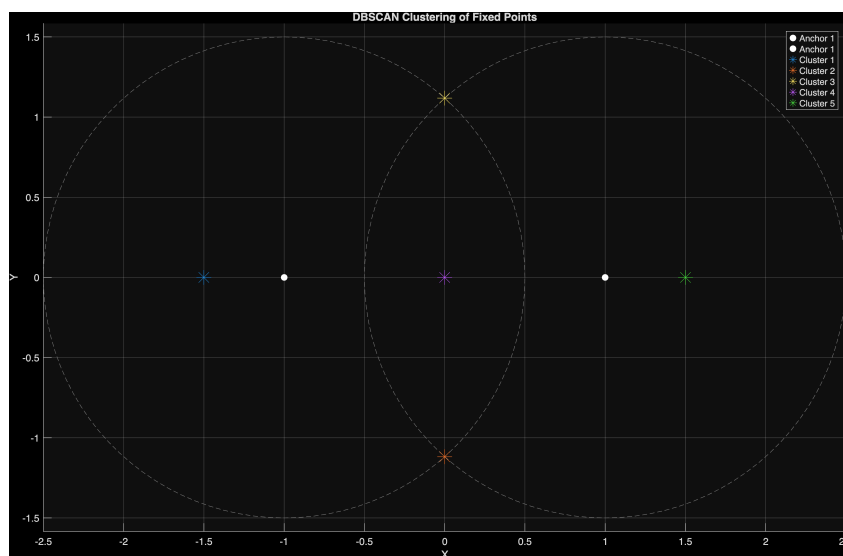


Figure 5: DBScan - Correct Identification

3.3. Fixed Points = 1

```
matlab
% System Constants
p.m = 1;
p.g = 10;

p.b1 = 1;
p.b2 = 1;;

p.d = 1; % drag coef

p.k1 = 10;
p.k2 = 10;
p.l1_0 = 1;
p.l2_0 = 1;

p.r1 = [-1; 0];
p.r2 = [1; 0];
```

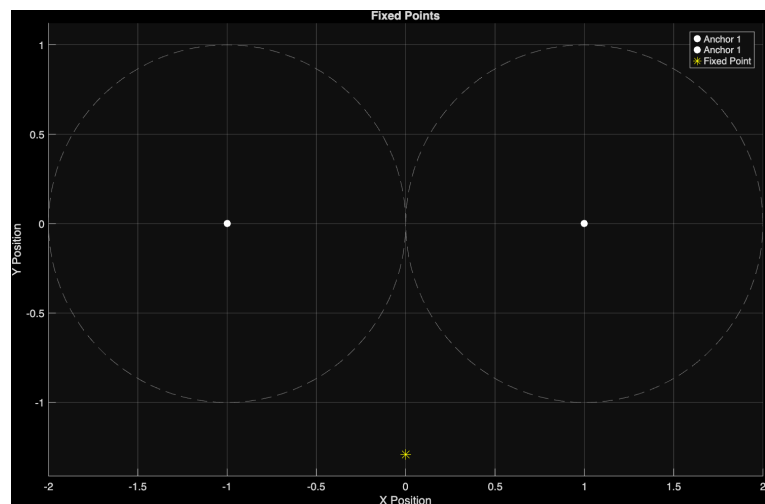


Figure 6: Fixed Points

Both KMeans and DBScan provided the right answer for this.

3.4. Fixed Points = 4, Two of them are very close

```
matlab
% System Constants
p.m = 1;
p.g = 0;

p.b1 = 1;
p.b2 = 1;;

p.d = 1; % drag coef

p.k1 = 10;
p.k2 = 10;
p.l1_0 = 1;
p.l2_0 = 1;

p.r1 = [-1; 0];
p.r2 = [1; 0];
```

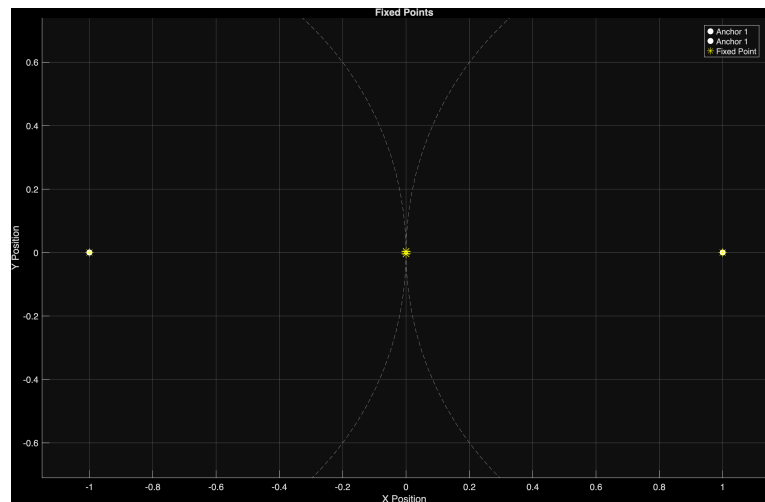


Figure 7: Fixed Points

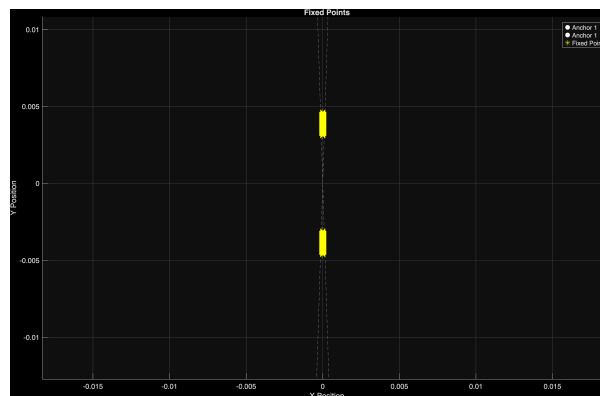


Figure 8: 2 Fixed Points Closeby

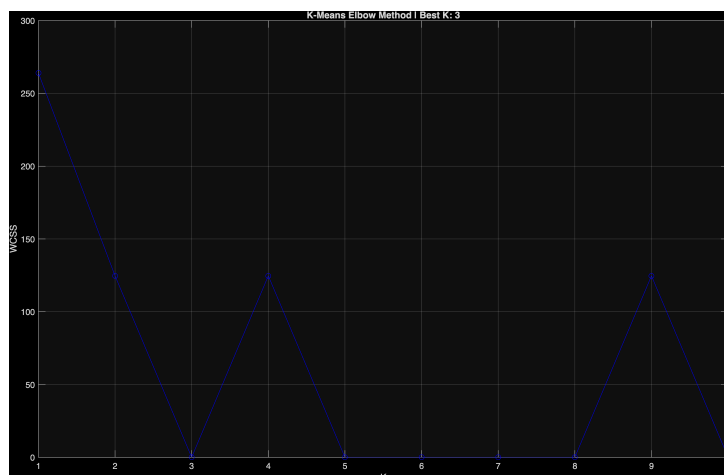


Figure 9: KMean - Best K = 3: WRONG

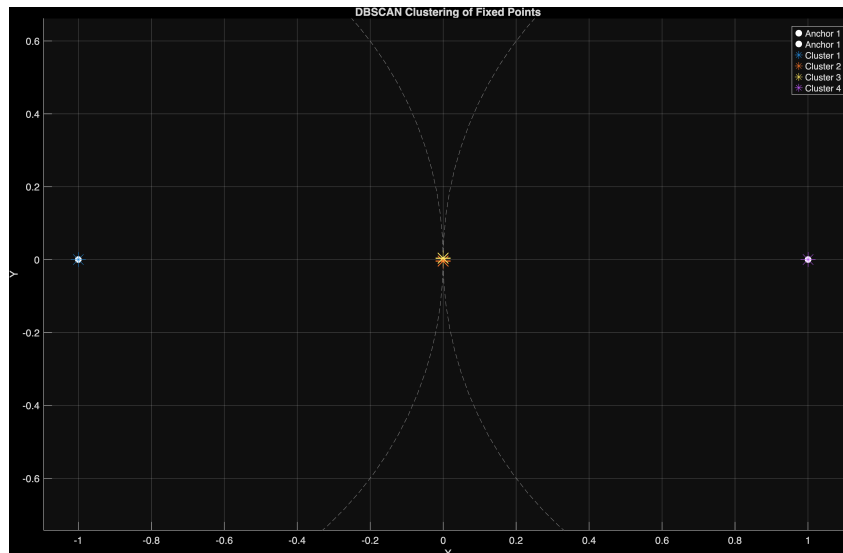


Figure 10: DBScan - Correct Identification

3.5. Fixed Points = 3

```
matlab
% System Constants
p.m = 1;
p.g = 1;

p.b1 = 1;
p.b2 = 1;

p.d = 1; % drag coef

p.k1 = 1;
p.k2 = 1;
p.l1_0 = 1;
p.l2_0 = 2;

p.r1 = [-1; 0];
p.r2 = [1; 0];
```

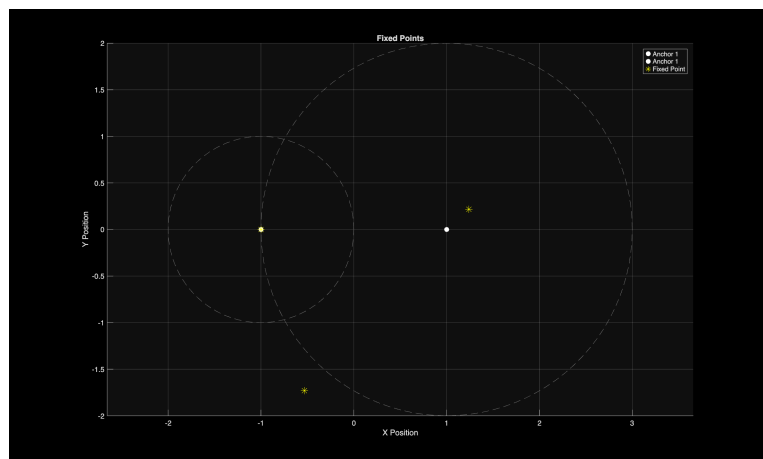


Figure 11: Fixed Points

DBScan was correct all the times, KMeans sometimes said best K was 2, while other times said it was 3.

3.6. Fixed Points = Infinite

When gravity is off, and either of the spring constants are zero, all points on the free length circle of the active spring are valid unique fixed points.

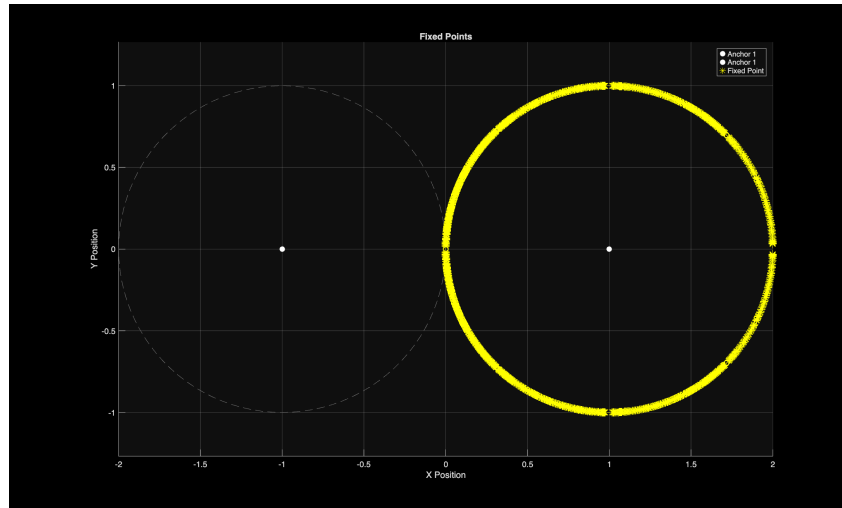


Figure 12: $g = 0, k_1 = 0 \Rightarrow \text{inf FP}$

Also when gravity is off, and both the springs are active, but the anchor positions are the same, we get infinite fixed points.

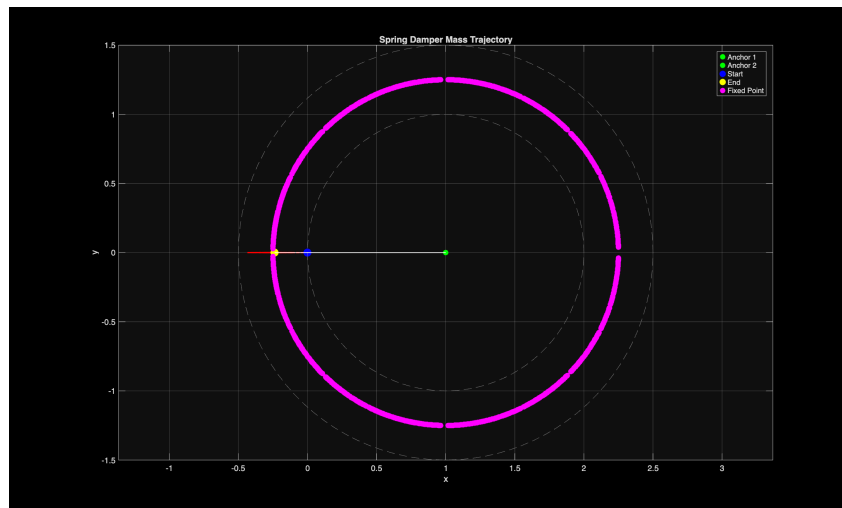


Figure 13: $g = 0, r_A = r_B \Rightarrow \text{inf FP}$

4. Linearised Motion

Task:

- Using the Jacobian supplied by fsolve at one of the stable fixed points, solve the linearized equations for motion for an initial condition near the fixed point.
- Using the same initial condition, use ODE45 to find the motion.
- Repeat for an unstable fixed point

System Used: The same parameters that resulted in 3 fixed points were used.

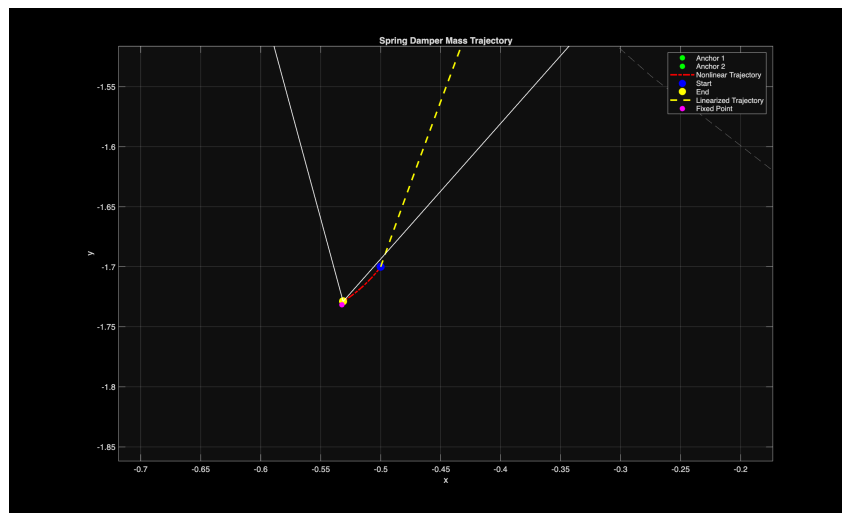
Stable Fixed Points:

- $[-1; 0]$
- $[-0.5323; -1.7316]$

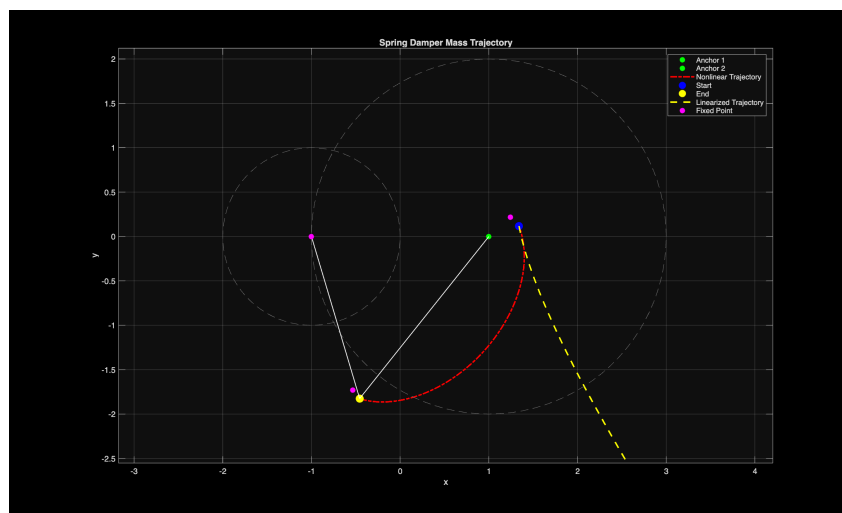
Unstable Fixed Points:

- $[1.2412; 0.2169]$

4.1. Response near stable fixed point



4.2. Response near unstable fixed point



4.3. Response When $L_0 = 0$, $b = 0$

